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Fine-grained multi-prompt essay scoring with multi-level disentanglement

Donghee Han[1] · Daeyoung Roh[1] · Euihwan Han[2] · Hwanjun Song[3] · Mun Yong Yi[3]

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Abstract
The application of language models in essay scoring has gained significant attention in recent years, typically on the basis of evaluating a single model across multiple prompts. However, in a multi-prompt setup, it is crucial to understand the varying aspects of different prompts. In such settings, there exist notable variations even in a trait with the same name across prompts. This semantic variation on the same traits underscores the need to treat them differently at a fine-grained level according to each prompt. In this study, we propose a multi-level disentanglement framework for multi-prompt essay scoring, designed to achieve fine-grained disentanglement of semantic differences across such traits. Our method not only improves the quality of the essay scoring, but also reduces memory usage and latency. Experimental results highlight that our framework surpasses seven state-of-the-art essay scoring methods and large language model(LLM)-based zero-shot and few-shot approaches, achieving the highest agreement with human essay ratings.

Keywords Essay scoring · Multi-level disentanglement · Transformer · Language models · Multi-prompt

1 Introduction

Fueled by raid advances in various language models, automated essay scoring (AES) has recently garnered significant attention. Existing approaches leveraging pre-trained bidirectional encoder representations from transformers(BERT)-based models have shown promising results for essay scoring, particularly for single-prompt and single-overall score setups (Yang et al. 2020; Wang et al. 2022; Jiang et al. 2023). However, these methods have not been effectively extended to multi-prompt or multi-trait setups, which require effective handling of diverse and complex evalu-

Extended author information available on the last page of the article

ation scenarios. Most studies in this domain have focused narrowly on single prompts or a single overall score, limiting their applicability. Furthermore, most of the existing multi-prompt or multi-trait methods primarily rely on shallow features, such as Part-Of-Speech(POS) tokens, while excluding rich semantic information (Mathias and Bhattacharyya 2020; Ridley et al. 2021; Kumar et al. 2022; Do et al. 2023). This approach fails to capitalize on recent advances in large language models (LLMs), which offer powerful semantic understanding and reasoning capabilities.

In real-world essay assessment, students are generally asked to write and get evaluated on a variety of essay prompts, and each prompt has a number of traits that reflect different aspects of the quality of the writing (Mathias and Bhattacharyya 2018). These traits serve as indicators that reflect various perspectives, including consistency, organization, grammar, and content relevance, while collectively providing a more comprehensive and fine-grained evaluation of a student's writing skills. Importantly, such trait-level feedback serves as valuable guidance for students in identifying and improving specific areas of their writing skills. Recent studies have aimed to exploit the interrelationships between these traits during model training, observing meaningful correlations that can improve overall evaluation performance (Do et al. 2023; Wang and Liu 2025). As a result, recent research efforts have increasingly focused on multi-prompt, multi-trait evaluation setups, which more accurately reflect the complexity and granularity of real-world educational assessment scenarios (Li and Ng 2024).

The latest approaches employ pre-trained Transformer models with an encoder-decoder structure to learn the relationships between traits in a multi-prompt and multi-trait essay scoring (Do et al. 2024). This model generates text and score in a sequence-to-sequence (Seq2Seq) manner, achieving better performance in multi-prompt and multi-trait essay scoring. In these studies, traits have been treated simply as equivalent if the names are identical on the prompts. However, this approach overlooks the fact that different prompts may have different evaluation criteria for traits or different semantic factors embedded in the essay that affect the score.

Figure 1 shows that even the same trait can contain different evaluation criteria depending on the prompt. Figure 2 illustrates the gaps in the embedding distributions of the same trait (i.e., *content*) between essay prompts. Even for essays with high scores on the same trait, the embedding distributions vary significantly depending

[Content Rubric of P1, P2, P8]
This property checks for the amount of content and ideas present in the essay.
Score 6: The writing is exceptionally clear, focused, and interesting. It holds the reader's attention throughout. Main ideas stand out and are developed by strong support and rich details suitable to audience and purpose. …

[Content Rubric of P5, P6]
Score 4: The response answers the question asked of it. Supporting evidence is specific to the memoir is used to support the points the writer makes. …

[Content Rubric of P7]
Score 3: Tells a story with ideas that are clearly focused on the topic and are thoroughly developed with specific, relevant details. …

Fig. 1 The evaluation criteria for *content* trait vary depending on the prompts

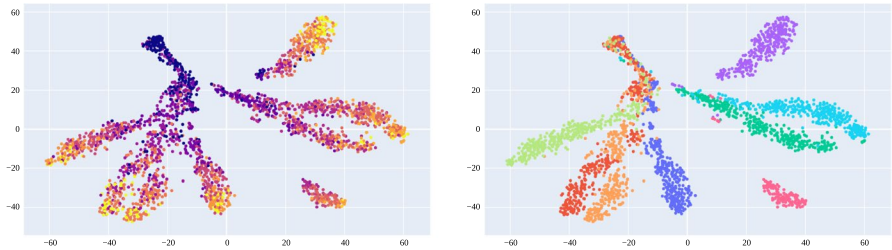

Fig. 2 T-SNE visualisation of the embeddings extracted from the language model trained to evaluate *content* scores. (Colors: Left-Score, Right-Prompt)

on the prompt. Hence, overlooking such differences and treating representations as if they share the same distribution over different prompts collapses prompt-specific information[1].

Although a recent study proposed prompt-wise disentanglement in multi-prompt essay scoring (Jiang et al. 2023), it still lacks granularity because it only evaluates the *overall* trait and overlooks the variations in how the same trait manifests across different prompts in its disentanglement phase. Therefore, this semantic variation over the same traits underscores the need to treat them differently at a fine-grained level according to each prompt. Furthermore, the state-of-the-art Seq2Seq essay scoring framework is not appropriate to achieve trait-level disentanglement. It turns out that there is a strong dependency between traits in scoring generation. The order of traits generated by auto-regressive decoding significantly affects the scoring performance of the framework (Do et al. 2024). Besides, the Seq2Seq fashion results in high latency due to the need to maintain a Transformer decoder in addition to the encoder, and its expensive inference cost of auto-regressive decoding.

In this paper, we propose a practical and effective multi-level disentanglement framework for multi-prompt essay scoring, MLPAS (**M**ulti-**L**evel **P**rompt-**A**ware Multi-trait Essay **S**coring), designed to achieve fine-grained disentanglement at both prompt- and trait-levels. In this framework, we introduce the notion of *token-based instruction*, a set of instruction tokens that are input to the Transformer encoder. The semantic embedding of the prompt text is extracted by a frozen text embedding model, followed by concatenation with a learnable prompt ID token. They are then concatenated with multiple learnable trait tokens as the final instruction input. It provides prompt-specific information about the target essay to guide the model in displaying the essay score for each trait token. Furthermore, we leverage *multi-level disentanglement*, which introduces prompt-wise projection and trait-wise predictions to select the most suitable parameters for the target prompt and traits. Here, the proposed method enables fine-grained disentanglement at both prompt- and trait-levels by adjusting the projection layers on demand.

While each component is simple, the merger of them facilitates fine-grained, multi-level disentanglement with an efficient encoder-only Transformer architecture.

[1] In our study context, the term *prompt* does not refer to the recent usage of instruction prompts with regard to LLMs, but rather to the written passages that tell students about what to write for their essays.

This advantage of our framework enables for a more accurate essay scoring with low latency. Our main contributions are as follows.

- We propose a novel multi-level disentanglement framework MLPAS for multi-prompt and multi-trait essay scoring, enabling fine-grained disentanglement at both prompt and trait levels.
- MLPAS demonstrates the highest agreement with human essay ratings compared to existing state-of-the-art multi-prompt essay scoring methods.
- MLPAS outperforms existing state-of-the-art essay scoring methods even in cross-prompt setups.
- We experimentally demonstrate that our method achieves superior performance while utilizing fewer parameters and exhibiting lower latency compared to the most recent method.

2 Related work

This section provides an overview of key research findings and trends in automated essay scoring that are closely related to our work.

2.1 Single-prompt essay scoring

Early research on AES focused primarily on prompt-specific essay scoring, training, and testing each model for each essay prompt (Tay et al. 2018; Dong et al. 2017; Uto et al. 2020). Deep learning techniques such as long short-term memory (LSTM) networks, bidirectional LSTM (BiLSTM) networks, and convolutional neural networks (CNNs) have been applied to the task of automated essay scoring. Recently, pre-trained Transformer models, such as BERT (Devlin et al. 2019), have been successfully applied to prompt-specific essay scoring, yielding significant improvements in overall scoring accuracy (Yang et al. 2020; Wang et al. 2022). Such pre-trained models leverage large corpora to capture rich semantic and syntactic information, making them excellent for a variety of tasks including essay scoring. For a more precise scoring, several studies have proposed using multiple traits in addition to the single overall score (Mathias and Bhattacharyya 2018, 2020; Hussein et al. 2020). Specifically, Kumar et al. (2022) proposed multi-task learning with word-level CNN architectures. However, it requires training separate models for each trait, implying substantial resource inefficiency.

2.2 Multi-prompt essay scoring

In real-world essay scoring, there are many types of essay prompts. Thus, it would be inefficient to create a separate model for each prompt. To address this issue, multi-prompt essay scoring models have been proposed that can evaluate essays from multiple prompts simultaneously. Multi-prompt essay scoring can be categorized into prompt-adaptive setting and cross-prompt setting.

The prompt-adaptive setup involves adjusting the scoring criteria based on the specific prompt to improve the accuracy of the essay evaluation (Mathias and Bhattacharyya 2020; Kumar et al. 2022). The most recent work (Do et al. 2024) uses a pre-trained Seq2Seq Transformer, thereby achieving higher performance in multi-prompt and multi-trait essay scoring. Nevertheless, they are inadequate for capturing the fine-grained level of semantic difference across prompts and traits, due to their text generation instability, with high latency, posing an extra challenge for efficient essay scoring.

The cross-prompt setup involves applying the evaluation model trained on seen prompts to unseen ones, thereby generalizing its effectiveness across various prompts. Prior studies conducted under this setup evaluated essays using a single overall score or multiple trait scores (Cao et al. 2020; Jiang et al. 2023; Do et al. 2023; Chen and Li 2023, 2024). In the most recent study, LLM was used to generate grammatically corrected data, to improve grammar-related evaluation metrics (Do et al. 2025).

Despite advancements in various language models, most multi-prompt and multi-trait essay scoring methods have not effectively utilized pre-trained language models capable of leveraging semantic information and context. This study aims to address these limitations by proposing an efficient approach that leverages pre-trained language models for multi-prompt and multi-trait essay scoring. Notably, we demonstrate that the proposed method is not only effective in prompt-adaptive settings, but also excels in cross-prompt settings, paving the way for a new direction in essay-scoring research.

2.3 LLM-based essay scoring

One emerging line of recent research focuses on LLM-based essay evaluation (Xiao et al. 2025; Seßler et al. 2025; Stahl et al. 2024). Open-source and closed-source LLMs are being explored for assessing student essays. These studies highlight the potential of such models to align with human raters, particularly in language-related aspects, and to reduce teachers' workload. However, they also reveal several limitations. For example, current LLMs may exhibit overestimation tendencies, face challenges in accurately assessing content quality, and still fall short of consistently outperforming state-of-the-art AES models, suggesting that further research is needed to provide reliable and nuanced essay assessment and feedback based on LLM. In this context, we emphasize the need for continued exploration of training relatively small models and optimizing them through various architectural approaches.

2.4 Seq2Seq-based eassay scoring

A prior study utilized the pre-trained encoder-decoder Transformer to generate trait scores in an auto-regressive manner (Do et al. 2024). In the study, only essay prompt ID and essay tokens were used as inputs to the Transformer encoder, and then the decoder generated a single text sequence consisting of trait and score pairs as:

$$\hat{S} = Dec(Enc([P_{\text{id}}|E])) \in \mathbb{R}^{v}. \tag{1}$$

Here, $\hat{S}$ denotes the generated text sequence, *Dec* denotes the Transformer decoder, and P_{id} denotes the essay prompt ID. This text sequence was then parsed to predict the final trait scores as:

$$\hat{Y} = g(\hat{S}) \in \mathbb{R}^n. \tag{2}$$

Here, g is a parser that extracts a scalar value corresponding to each trait score from the text sequence.

This method outperformed existing other essay scoring methods, but overlooked the representation disentanglement, and also suffered from high latency and additional parameters due to its auto-regressive nature. To address these limitations, we propose an encoder-only model that facilitates fine-grained disentanglement at both prompt- and trait-levels and replaces the complicated auto-regressive task with a simple regression task for high effectiveness and efficiency.

3 Method

In this section, we provide the overview of our proposed method and detailed explanations of its components. Figure 3 presents an overview of our proposed method, and Table 1 offers detailed descriptions of the variables employed.

3.1 Overview

We adopt an encoder-only architecture combined with multi-level disentanglement to achieve better performance with reduced latency and fewer parameters. We use a pre-trained Transformer encoder to extract trait-wise features from prompts and essays. The three token sets are concatenated into an integrated sequence S.

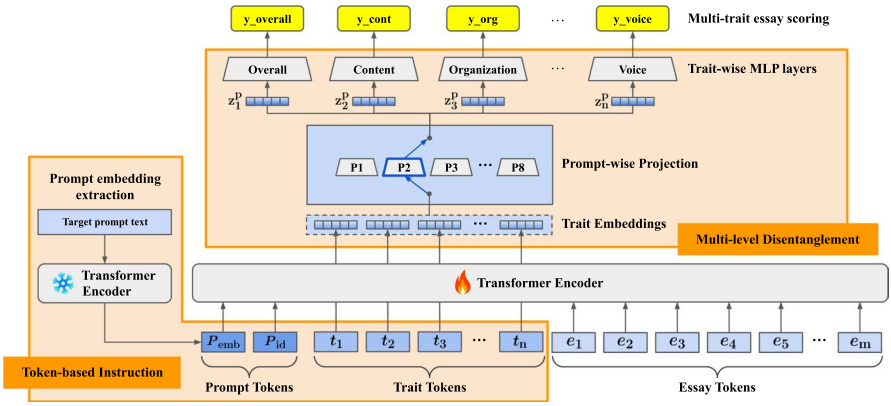

Fig. 3 Overview of MLPAS: Token-based instruction provides prompt-specific information about the target essay to guide the model in outputting the essay score for each trait token, while multi-level disentanglement dynamically adjusts the projection layers for fine-grained, multi-level disentanglement over prompts and traits

Table 1 Variable definitions and their meanings

Variable	Meaning
S^*	Integrated sequence of concatenated prompt, trait, and essay token embeddings
P^*	Concatenated prompt embedding and prompt ID token embedding
Z	Representation of the concatenated sequence after encoding
Z_T	Extracted trait token embeddings
Z_T^p	Prompt-specific projection of trait embeddings
T	Sequence of trait tokens
E	Sequence of essay tokens
t_i	*i*-th trait token
e_i	*i*-th essay token
p_{emb}	Prompt embedding
p_{id}	Prompt ID token
W^p	Weights for prompt-wise projection for prompt *p*
b^p	Biases for prompt-wise projection for prompt *p*
MLP_t	Multi-layer perceptron for trait *t* score prediction
z_t^p	Final embedding of trait *t* in prompt *p*
y_t	Predicted score of trait *t*

$$S = [P^*|T|E] \in \mathbb{R}^{(2+n+m)\times d}. \tag{3}$$

Here, P^* represents the concatenated prompt embedding and the embedding of the prompt ID token.

The Transformer encoder processes *S* to generate the representation *Z*:

$$Z = \mathrm{Enc}(S) \in \mathbb{R}^{(2+n+m)\times d}. \tag{4}$$

The representation *Z*, which integrates the prompt, trait tokens, and essay, enables efficient multi-trait score prediction by passing the representation through a non-auto-regressive prediction function *f*, denoted as:

$$\hat{Y} = f(Z) \in \mathbb{R}^{n\times d}. \tag{5}$$

This process enables the model to generate predictions for the target traits based on the information extracted from the prompt, essay, and target traits. Unlike the previous method described in Eqs. (1) and (2), we do not follow the auto-regressive approach to predict final scores, which then allows us to predict each trait in parallel via regression.

3.2 Main components

In this subsection, we provide a detailed explanation of each component of our proposed method.

3.2.1 Token-based instruction

Prior studies often combine essay text and prompt IDs into a single sequence (Do et al. 2024), but this approach struggles to use the semantic information of the prompts and to learn the distinct representations specific to the traits. To address these limitations, we introduce three types of instruction tokens. The first type reflects the semantic information contained in essay prompt through prompt embedding, the second type refers to special tokens assigned according to the prompt ID, and the third type consists of trait tokens, crucial for capturing trait-specific information. We assign a special token to each trait, and if the trait is not relevant to the prompt, it is masked with an *NA* token.

This design enables the model to leverage the semantic information and specific instructions embedded within each essay prompt. To efficiently incorporate prompt information without significantly increasing input length or requiring additional encoder training, we convert each prompt into a single token embedding, which is then inputted as a single token. Additionally, for prompt ID, we assign a special token for each prompt ID, allowing their embeddings to be learned and updated during training. This approach facilitates the model's ability to capture prompt-specific distinctions, thus enhancing its capacity to generalize across diverse prompts during the learning process.

The fixed pre-trained Transformer encoder and average pooling are used to obtain prompt embedding $p_{\text{emb}} \in \mathbb{R}^d$ as:

$$p_{\text{emb}} = \text{avg}(\text{Enc}^*(P)). \tag{6}$$

Here, Enc^* denotes a fixed encoder from pre-trained Transformers and P denotes essay prompt text of target essay. In our implementation, we used the same backbone model for Enc^* and Enc (e.g. t5-base). The prompt embedding and the prompt ID token together form P^* as:

$$P^* = [p_{\text{emb}} \| p_{\text{id}}] \in \mathbb{R}^{2\times d}. \tag{7}$$

These tokens enable the model to recognize and process information differently depending on the prompt. This approach can be applied regardless of the length of the prompt because it needs a single extracted token, and has the advantage of being able to extract more fine-grained information by calculating the attentions with the prompt embedding at the essay text token level.

The final input sequence is formalized as:

$$S = [P^* | T | E] = [\underbrace{p_{\text{emb}}, p_{\text{id}}}_{\text{Prompt Tokens}}, \underbrace{t_1, t_2, \ldots, t_n}_{\text{Trait Tokens}}, \underbrace{e_1, e_2, \ldots, e_m}_{\text{Essay Tokens}}]. \tag{8}$$

This input sequence is fed into the Transformer encoder to obtain the sequence representation. This token-based instruction is illustrated on Fig. 3.

3.2.2 Multi-level disentanglement

We design a multi-level disentanglement approach to predict trait scores from trait token embeddings[2] to consider prompt-wise, trait-wise differentiation. We dynamically select the projection layer for multi-level disentanglement. It involves a two-step process: (1) *Prompt-wise projection* and (2) *Trait-wise multi-layer perceptrons (MLPs)*.

Firstly, we extract the trait token embeddings from the sequence embedding Z. Let $Z_T \in \mathbb{R}^{n \times d}$ represent the extracted trait token embeddings and $z_i \in \mathbb{R}^d$ represent the embedding of trait t_i:

$$Z_T = [z_1, z_2, \ldots, z_n]. \tag{9}$$

Prompt-wise projection captures the distinct characteristics of each prompt within the embeddings of the traits. Then, for each prompt, unique weights and biases are utilized:

$$Z_T^p = W^p \cdot Z_T + b^p. \tag{10}$$

Here, $W^p \in \mathbb{R}^{d \times d}$ and $b^p \in \mathbb{R}^d$ represent the weights and biases to project a specific prompt p, a set of specified projection parameters are dynamically selected based on the given prompts.[3]

Trait-wise MLPs predict the scores for each trait from the adapted trait embeddings:

$$\hat{y}_t = \mathrm{MLP}_t(z_t^p). \tag{11}$$

In the equation, MLP_t represents the multilayer perceptron used to predict the score of the trait t, z_t^p is the adapted embedding of the trait t, and $\hat{y}_t$ is the predicted score of the trait t.

These components capture trait variability across prompts by addressing both prompt-specific and trait-specific differences, leading to superior performance in multi-prompt and multi-trait essay scoring. This process is illustrated on Fig. 3.

3.3 Model training

We train our model in a multi-prompt, multi-trait setting, consistent with previous studies (Mathias and Bhattacharyya 2020; Kumar et al. 2022; Do et al. 2024). The training set comprises a mixture of essays from various prompts, each evaluated on multiple traits.

We update the model by calculating the Mean Squared Error (MSE) loss for each trait, masking the non-existent traits during the process:

[2] The subset of the trait tokens in Eq. (9) from the Transformer encoder's output in Eq. (4).

[3] Similar to the recent work, we assumes that prompt ID is given, while it can be extended to the cross-prompt setting by averaging the projections of trained prompts.

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\sum_{t=1}^{T} m_{i,t}}\sum_{t=1}^{T} m_{i,t}(\hat{y}_{i,t} - y_{i,t})^2. \tag{12}$$

N is the total number of samples and T is the total number of traits. For each sample i and trait t, $\hat{y}_{i,t}$ denotes the predicted score; $y_{i,t}$ the true score. The mask $m_{i,t}$ is 1 if trait t exists for sample i and 0 otherwise. This loss function ensures that only the traits present for each sample contribute to the loss, allowing the model to learn effectively from the data with differences in the set of traits between the prompts. We normalize scores to a range of 0 to 10 for consistency across prompts and traits. For evaluation using the Quadratic Weighted Kappa (QWK) metric, we then map these scores back to their original trait-specific ranges.

3.4 Applying to cross-prompt setup

In this subsection, we explain how to extend the proposed method to cross-prompt setups. By leveraging the prompt-wise projection and normalization techniques, MLPAS can be adapted to handle unseen prompts during inference. We apply prompt-wise z-score normalization to the embeddings produced by the prompt-wise projection. Specifically, given the embedding Z_T^p for a specific prompt p, we normalize it using:

$$\tilde{Z}_T^p = \frac{Z_T^p - \mu_p}{\sigma_p} \tag{13}$$

where μ_p and σ_p represent the mean and standard deviation of Z_T^p for prompt p, respectively. $\tilde{Z}_T^p$ is the normalized embedding for the given prompt p after normalization. This normalization ensures that the embeddings are standardized, mitigating the distribution gap between prompts and risk of overfitting to the training dataset.

The trait-wise MLP layers are then trained using these normalized embeddings to predict the trait scores. At this stage, the input embeddings are fixed and only the trait-wise MLP layers are updated.

$$\hat{y}_t = \text{MLP}_t(\tilde{Z}_T^p) \tag{14}$$

where $\hat{y}_t$ denotes the predicted score for trait t, MLP_t represents the multi-layer perceptron used for predicting the score of trait t.

During inference, we assume that the target prompt ID is unknown, we use average pooling across all the trained prompt-wise projections to generate the embeddings. The average pooled embedding is computed as:

$$\overline{Z}_T = \frac{1}{K}\sum_{p=1}^{K} Z_T^p \tag{15}$$

where K is the number of trained prompts and $\overline{Z}_T$ is the generalized embedding for unseen prompt sample. This pooling approach allows the model to make generalized predictions even when faced with unseen prompts.

4 Evaluation

In this section, we describe the experimental setup used to validate the effectiveness of our proposed method, present the results, and provide an in-depth analysis.

4.1 Dataset

We use the well-known Automated Student Assessment Prize (ASAP) and ASAP++ datasets (Mathias and Bhattacharyya 2018), which represent a widely adopted benchmark in the field of automated essay grading (Ke and Ng 2019). This extensively used dataset comprises approximately 13,000 English essays written by American high school students in grades 7 through 10, responding to eight distinct prompts. Each essay in the dataset was initially evaluated by two human raters. However, a limitation of the original ASAP dataset is that only two of the eight prompts include scores for individual essay attributes such as content, organization, and style, with the remaining six primarily providing overall holistic scores. To overcome this limitation and leverage the evaluation trait scores across all prompts in our research, we combined the original ASAP dataset and the ASAP++ dataset, effectively merging these resources. This combined dataset allows us to analyze and model individual writing traits for a wider range of essays, while still leveraging the resources that have been the basis of numerous related studies within the automated essay scoring community. Consistent with previous studies (Kumar et al. 2022; Do et al. 2024), we used the same source data and ran our experiments using the same 5-fold split.

To further ensure the reliability of our evaluation, we also utilized the Feedback Prize dataset,[4] which comprises argumentative essays written by 8th-12th grade english language learners. These essays have been scored on six analytical measures: cohesion, syntax, vocabulary, phraseology, grammar, and conventions, each representing a component of essay writing proficiency, with scores ranging from 1.0 to 5.0 in increments of 0.5. Although the original dataset includes essays written in response to 44 different prompts, all essays were evaluated using the same trait set; to prevent excessive data fragmentation at the prompt level, we grouped the prompts into three broader categories based on thematic relatedness: Education and School Life, Personal Growth and Attitude, and Social Experiences and Future. Table 4 presents the grouping of original essay prompts based on the thematic categories we manually defined. We also performed 5-fold cross-validation on this dataset. Tables 2 and 3 show the compositions of the datasets, revealing significant overlaps in traits across distinct domains (Table 4).

[4] https://www.kaggle.com/competitions/feedback-prize-english-language-learning/data.

Table 4 Manually grouped essay prompts by thematic category in the Feedback Prize dataset

Thematic group	Original essay prompts
Education and School Life	Distance learning, Grades for extracurricular activities, Lunch menus, Lessons with elementary school students, Year-round school, Cell phones in classrooms, Three-year high school program, Controlling extracurricular involvement, Creative arts requirement, Afterschool homework club, Mandatory extracurricular activities, Extended school day, Enjoyable educational activities
Personal Growth and Attitude	Benefits of a problem, Individuality, Career commitment, Positive attitudes, Seeking multiple opinions, Benefits of a good attitude, Imagination, Self-reliance, Honesty, Being busy, Success and failure, Trying something beyond what you have mastered, Praising student work, First impressions, Setting our aim, Influences of character, Talents and skills
Social Experiences and Future	Community service, Impact of technology, Influencing behavior, Curfews for teenagers, Four-day work week, Places to visit, Cell phones at school, Learning from the experience of others, Letter to employer, Future accomplishments, Working with a group or alone, Summer projects, Spending time outdoors, Internships and shadowing

Table 2 Overview of ASAP/ASAP++ dataset used in experiments

Prompt	Essay domain	# Essays	Avg. length	Available traits
1	Computer usage	1,785	350	Over, Cont, WC, Org, SF, Conv
2	Library censorship	1,800	350	Over, Cont, WC, Org, SF, Conv
3	Cyclist setting	1,726	150	Over, Cont, PA, Nar, Lang
4	Story analysis	1,772	150	Over, Cont, PA, Nar, Lang
5	Memoir mood	1,805	150	Over, Cont, PA, Nar, Lang
6	Empire state building	1,800	150	Over, Cont, PA, Nar, Lang
7	Patience story	1,569	150	Over, Cont, Org, Conv, Style
8	Laughter importance	723	150	Over, Cont, WC, Org, SF, Conv, Voice

Over: Overall, Cont: Content, WC: Word Choice, Org: Organization, SF: Sentence Fluency, Conv: Conventions, PA: Prompt Adherence, Nar: Narrativity, Lang: Language

Table 3 Overview of feedback prize dataset used in experiments

Prompt	Essay domain	# Essays	Avg. length	Available traits
1	Education and school life	1,095	431.4	Coh, Syn, Voc, Phr, Gram, Conv
2	Personal growth and attitude	1,865	435.9	Coh, Syn, Voc, Phr, Gram, Conv
3	Social experiences and future	945	418.8	Coh, Syn, Voc, Phr, Gram, Conv

Conv: Conventions, Gram: Grammar, Phr: Phraseology, Voc: Vocabulary, Syn: Syntax, Coh: Cohesion

4.2 Compared baselines

We compare the performance of our model with these baselines to demonstrate the effectiveness of our model in multi-prompt and multi-trait essay scoring.

4.2.1 Baselines for prompt-adaptive setup

We evaluate our model against several existing prompt-adaptive essay scoring methods in multi-prompt, multi-trait settings.

- **STL-LSTM** (single-task learning with LSTM) (Dong et al. 2017) is an LSTM-CNN-based model for scoring essays in single task learning scenarios.
- **HISK** (histogram intersection string kernel) (Cozma et al. 2018) utilizes a histogram intersection string kernel with a support vector regressor.
- **MTL-BiLSTM** (multi-task learning with BiLSTM) (Kumar et al. 2022) employs a BiLSTM architecture for scoring essays in multitask learning sets.
- **ArTS** (autoregressive prediction of multi-trait scores) (Do et al. 2024) is our main baseline, a recently proposed T5-based encoder-decoder model specifically designed for automated essay scoring and is considered a top-performing baseline.

4.2.2 LLM-based baselines for prompt-adaptive setup

To further benchmark our method, LLM-based baselines using GPT-4[5] and GPT-4o[6] under zero-shot and few-shot settings are compared. In these baselines, randomly sampled essays with their corresponding scores are provided as demonstration examples within the same prompt for zero-shot, 3-shot and 5-shot configurations. Here is the prompt sample used in our experiment to evaluate LLM's essay scoring capability. The prompts included the essay scoring instruction, the essay text, and the specific traits, along with their respective maximum and minimum scores. Below is an example of the zero-shot prompt:

[5] https://openai.com/index/gpt-4/

[6] https://openai.com/index/hello-gpt-4o/

LLM Essay Scoring Prompt

Evaluate the following essay based on the given traits and their score ranges. Provide only numeric scores without explanation. Format the scores as 'Trait: Score' for each trait.

"Patience, whats the first word that comes to your mind when you hear that word? Waiting? I know that's the main word in my mind. Here is a story when I was very patient. Every kid dreads meap testing it was @DATE1 and we have to be completely silent. It's hard enough to be quite. But it's harder when your next to your friends, and your a girl. We had to do meap writing, writing is the most wrost for me. But when I was done I felt really good and quiet for everyone else. That's the story when I had to be patient."

Overall: 0 to 30
Content: 0 to 6
Organization: 0 to 6
Conventions: 0 to 6
Style: 0 to 6

We also compared our approach with ArTS-LLaMA2, a fine-tuned version of the LLaMA2-13B decoder-only model proposed in prior work (Do et al. 2024), which was designed to investigate the effectiveness of autoregressive score generation for AES.

4.2.3 Baselines for cross-prompt setup

We evaluate the effectiveness of our proposed method in the cross-prompt setup by comparing it with the state-of-the-art cross-prompt AES models. All these baselines are models designed to predict multiple traits in a prompt cross-prompt setting and were known to achieve the highest performance prior to our study.

- **CTS** (Cross-prompt Trait Scorer) (Ridley et al. 2021) is a cross-prompt automated essay scoring method designed to predict overall scores for unseen-prompt essays, addressing the challenge of generalization in AES tasks.
- **PMAES** (prompt representation learning framework for cross-prompt AES) (Ridley et al. 2020) is a cross-prompt automated essay scoring method that introduces a prompt mapping strategy to align the source and target prompts, allowing the extraction of more consistent shared features for better essay representation and scoring.
- **PLAES** (Prompt-generalized and Level-aware learning framework for crossprompt Automated Essay Scoring) (Chen and Li 2024) is a cross-prompt automated essay scoring framework that employs meta-learning to acquire generic knowledge across source prompts and a level-aware learning strategy, incorporating general and level-specific scorers, to improve writing-level differentiation and scoring performance through contrastive learning techniques.
- **GAPS** (grammar-aware cross-prompt trait scoring) (Do et al. 2025) is a grammar-aware cross-prompt trait scoring method that leverages grammar error correction to capture prompt-independent syntactic aspects for generic essay representation. By internally referencing both original and corrected essays, GAPS focuses on generalizable features, achieving significant improvements in prompt-independent and grammar-related trait scoring, particularly in challenging cross-prompt scenarios. However, as an LLM-augmented approach that uses grammar error

correction, GAPS is likely to require substantial computational resources.

4.2.4 Evaluation metric

We leverage the quadratic weighted kappa (QWK), a widely adopted metric in existing AES studies (Ke and Ng 2019; Ramesh and Sanampudi 2022). QWK is known for its effectiveness in capturing agreement between human-rated and model-predicted scores. We report the average QWK of the trained model, which performs best in the validation dataset. To ensure a comprehensive evaluation, we report the aggregated QWK scores for each trait, i.e., trait-wise comparison, and for each prompt, i.e., prompt-wise comparison.

4.3 Main results

In this subsection, we present the main experimental results and verify the effectiveness of our proposed method in comparison to existing models.

4.3.1 Performance in prompt-adaptive setting

Tables 5 and 6 present the results of the agreement between automated essay evaluators and human ratings at the trait-wise level and the prompt-wise level, respectively, based on experiments conducted on the ASAP/ASAP++ dataset. Overall, the results show that MLPAS surpasses other models in most cases (9 out of 11 for trait-wise and 6 out of 8 for prompt-wise). Although it does not achieve the highest score in some cases (2 for traitwise and 2 for promptwise), the performance gap is marginal.

We particularly observe a significant improvement in P8 (the minority with respect to the number of training examples).[7] This improvement is mainly attributed to the robustness of MLPAS against bias from majority prompts (P1-P7), caused by their traits that overlap with those of P8. In the case of "Voice", it is not directly affected by the main prompts, but may be indirectly influenced by the overlapping traits of P8 (e.g., "Content" in P1 influencing "Content" in P8, which in turn affects "Voice" in P8). Our proposed method effectively reduces inter-prompt and inter-trait interference, demonstrating that we can achieve performance improvements in these more subtle prompts and traits. In sum, our multi-level disentanglement successfully captures the distinct characteristics of each trait at a fine-granular level, minimizing the collapse of each trait by other prompts. In addition, MLPAS outperforms ArTS, a Transformer model using the encoder-decoder structure, even though we eliminate the decoder part of the Transformer. Although ArTS does not perform well on the "Overall" trait, which is the most important among all traits, MLPAS exhibits the best agreement on that trait, highlighting the potential to use the encoder-only structure compared to the encoder-decoder structure.

To further validate the effectiveness of the proposed method, we conducted additional experiments on the Feedback Prize dataset. Table 9 presents the results of these

[7] On P8, the performance increase by 0.08 both for the non-overlapping trait (e.g., "Voice") and for traits that overlap with those in the major prompts.

Table 5 Trait-wise agreement with human ratings on the ASAP/ASAP++ dataset.

Model type	Model	Overall	Content	PA	Lang	Nar	Org	Conv	WC	SF	Style	Voice	AVG↑(SD↓)
	HISK	0.718	0.679	0.697	0.605	0.659	0.610	0.527	0.579	0.553	0.609	0.489	0.611 (-)
Non-transformer	STL-LSTM	0.750	0.707	0.731	0.640	0.699	0.649	0.605	0.621	0.612	0.659	0.544	0.656 (-)
	MTL-BiLSTM	0.764	0.685	0.701	0.604	0.668	0.615	0.560	0.615	0.598	0.632	0.582	0.638 (-)
Encoder-Decoder	ArTS	0.754	0.730	**0.751**	**0.698**	0.725	0.672	0.668	0.679	0.678	**0.721**	0.570	0.695 (±0.018)
Encoder-only	MLPAS **(ours)**	**0.771**	**0.746**	**0.751**	0.691	**0.732**	**0.710**	**0.701**	**0.701**	**0.699**	0.702	**0.650**	**0.714** (±0.007)

PA: Prompt Adherence, Lang: Language, Nar: Narrativity, Org: Organization, Conv: Conventions, WC: Word Choice, SF: Sentence Fluency

experiments. Our proposed method demonstrated significantly superior performance compared to existing approaches. Considering that the essays in the Feedback Prize dataset are, on average, longer than those in the ASAP/ASAP++ dataset, these results suggest that our approach is particularly effective at handling essays with richer and more extensive content.

4.3.2 Comparison with LLM-based methods

An interesting idea is to compare MLPAS with recent LLMs. The foundational model, such as GPT-4 and GPT-4o, is capable of performing a human-like evaluation with prompt tuning (Kojima et al. 2022; Liusie et al. 2024; Zhang et al. 2024).

Our finding is that the naive use of LLMs is inappropriate for accurate essay scoring due to the unclear scoring standards.

More specifically, we provide relevant traits and ranges of scores as prompts to LLMs for automatic essay scoring (see Appendix for the prompt details). Figure 4 illustrates the comparison of the performance of the zero-shot and few-shot evaluation of GPT-4 and GPT-4o. The results indicate that recent LLMs perform poorly compared to our method, highlighting the necessity of fine-tuning with clearly defined criteria for scoring.

Remarkably, our method outperformed ArTS-LLaMA2, despite ArTS-LLaMA2 having nearly 10 times more parameters. These findings suggest that, in the context of essay scoring, when sufficient training data is available, lightweight models can achieve competitive and even superior performance with significantly improved efficiency.

4.3.3 Performance in cross-prompt setting

We conducted cross-prompt experiments to evaluate the generalizability of our method to unseen prompts. We compare the performance of MLPAS with state-of-the-art cross-prompt AES models. As in previous studies, we trained the model on all prompts except the target prompt. To generalize to unobserved prompts in training, we applied prompt-wise z-score normalization to the embeddings from prompt-wise projection. We trained the trait-wise MLP layers with fix the normalized embeddings, to avoid overfitting to the training dataset. In the inference step, we assumed that the target prompt ID was unknown and, therefore, used average pooled embeddings utilizing all the trained prompt-wise projections.

As illustrated in Fig. 5, our method outperformed recent cross-prompt AES models both in mean score and in the majority of prompts and traits (6 out of 9 for traits and 6 out of 8 for prompts). Our approach leverages pre-trained language models, offering a clear advantage over existing POS tagging-based methods by utilizing the latest advancements in language model technology.

4.3.4 Model size and latency

A crucial aspect when using Transformers is scalability, achieved by replacing the backbone with a larger one. However, the increasing latency of the larger model is the main

Table 6 Prompt-wise agreement with human ratings on the ASAP/ASAP++ dataset (P1-P8 denote the prompts)

Model type	Model	P1	P2	P3	P4	P5	P6	P7	P8	AVG↑(SD↓)
	HISK	0.674	0.586	0.651	0.681	0.693	0.709	0.641	0.516	0.644 (-)
Non-transformer	STL-LSTM	0.690	0.622	0.663	0.729	0.719	0.753	0.704	0.592	0.684 (-)
	MTL-BiLSTM	0.670	0.611	0.647	0.708	0.704	0.712	0.684	0.581	0.665 (-)
Encoder-Decoder	ArTS	0.708	0.706	**0.704**	0.767	0.723	**0.776**	0.749	0.603	0.717 (±0.025)
Encoder-only	MLPAS **(ours)**	**0.718**	**0.726**	0.703	**0.771**	**0.727**	0.765	**0.753**	**0.683**	**0.731** (±0.007)

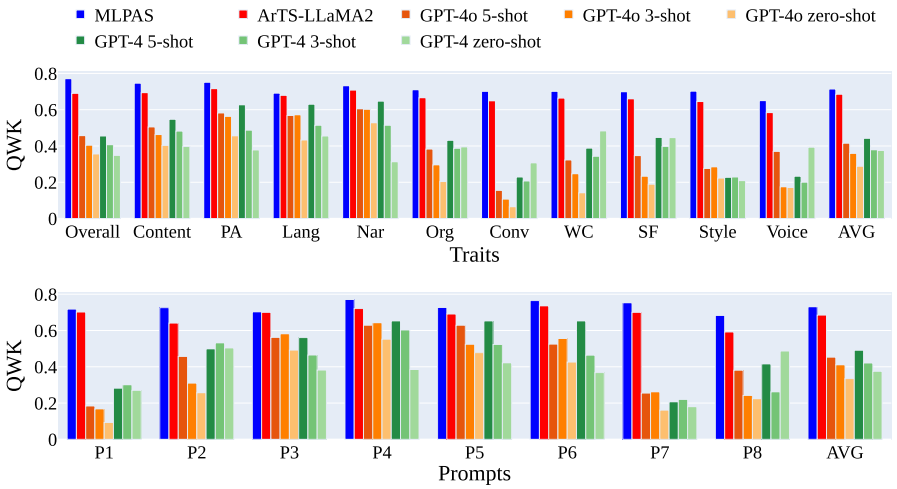

Fig. 4 Agreement with human ratings on the ASAP/ASAP++ dataset using LLMs as essay evaluators

bottleneck to hinder its practical use for online essay evaluation. Hence, we conducted an in-depth study on the trade-off between model size and latency. All inference time measurements were conducted using an RTX 3090 GPU with 24GB of VRAM. We fixed the batch size and measured the time required to process 100 randomly sampled test instances. The inference time reported represents the average processing time per essay.

Figure 6 illustrates the trade-off between model size and latency (in milliseconds) on three different sizes of T5 backbones, namely small, base and large from left to

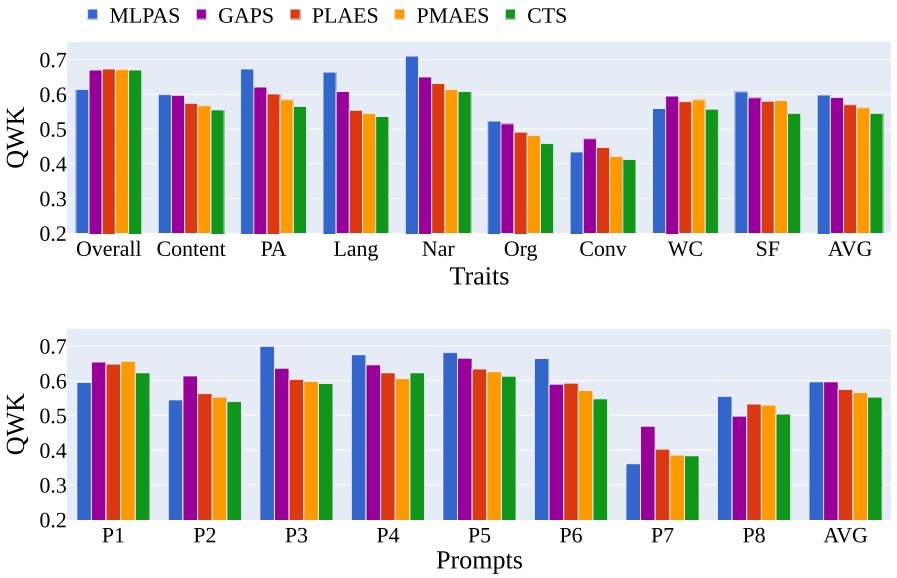

Fig. 5 Cross-prompt performance comparison on the ASAP/ASAP++ dataset

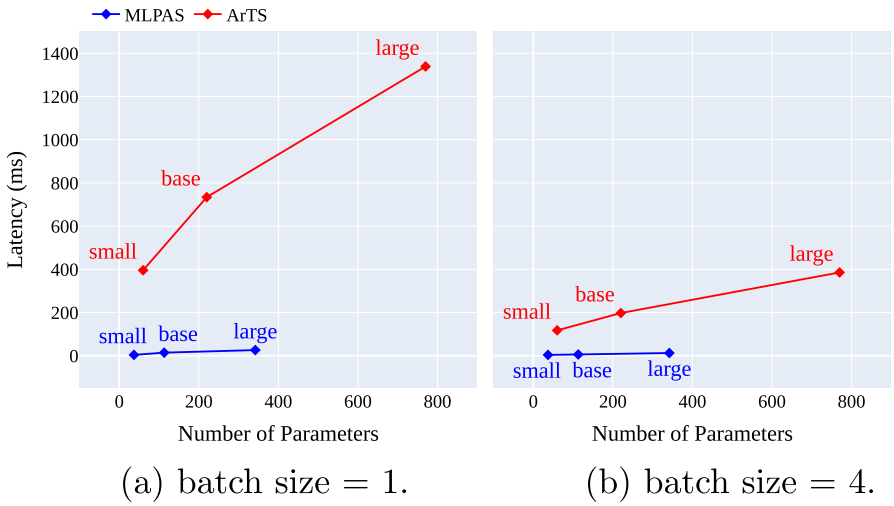

Fig. 6 Trade-off between model size (# parameters) and latency (ms) when using two different batch sizes

right. In particular, MLPAS exhibits a much better trade-off between them, achieving faster inference speed (lower latency) even with larger T5 backbones. This improvement is attributed to (1) the removal of the Transformer decoder and (2) the replacement of a computationally heavy autoregressive decoding task with a simple and efficient regression task.

Tables 7 and 8 present the QWK scores of ArTS and MLPAS, from three different backbone sizes. In general, both methods show good scalability with regard to the model size, considering that they achieve higher performance as the size of models gets larger (see the last "AVG" column for the tables). However, our framework MLPAS has better scalability compared to ArTS: (1) MLPAS (w. T5-small) achieves the average QWK score even better than ArTS (w. T5-large); (2) MLPAS (w. T5-large) is even faster than ArTS (w. T5-small) due to our efficient framework design, i.e., the encoder-only Transformer; and (3) MLPAS needs much fewer trainable parameters than ArTS, thus leading to less computational cost. These advantages over ArTS suggest that our model offers substantial advantages in real-world use cases (Table 9).

4.4 In-depth analysis

In this section, we present experiments and results that provide a more detailed analysis of each component of the model. We also conduct an in-depth analysis of our proposed method through visualizations.

4.4.1 Impact of multi-level disentanglement

We investigate the impact of multi-level disentanglement by MLPAS on traits with the same name across different prompts. Figure 7 visualizes the embedding of the trait just before the regression layer with and without our prompt projection. We observe a

Table 7 Trait-wise agreement with human scores on ASAP/ASAP++ dataset when using different sizes of T5 backbones

Model	# Parameters	Overall	Content	PA	Lang	Nar	Org	Conv	WC	SF	Style	Voice	AVG↑(SD↓)
ArTS (t5-small)	60 M	0.712	0.695	0.720	0.667	0.711	0.630	0.606	0.631	0.625	0.694	0.474	0.651 (±0.026)
ArTS (t5-base)	220 M	0.754	0.730	0.751	0.698	0.725	0.672	0.668	0.679	0.678	0.721	0.570	0.695 (±0.018)
ArTS (t5-large)	770 M	0.751	0.730	0.750	0.701	0.728	0.675	0.682	0.680	0.680	0.715	0.603	0.700 (±0.024)
MLPAS(T5-small)	37 M	0.760	0.733	0.738	0.681	0.719	0.690	0.686	0.696	0.689	0.711	0.640	0.704 (±0.007)
MLPAS(T5-base)	113 M	0.771	0.746	0.751	0.691	0.732	0.710	0.701	0.701	0.699	0.702	0.650	0.714 (±0.007)
MLPAS(T5-large)	342 M	0.768	0.752	0.760	0.700	0.731	0.719	0.709	0.702	0.713	0.693	0.641	0.717 (±0.008)

distinct difference with and without multi-level disentanglement by MLPAS. Without the prompt-wise projection, the unique characteristics of the trait in each prompt are likely to collapse (i.e., the overlap of trait embeddings across prompts). In contrast, the trait embeddings produced by MLPAS exhibit a clear separation across prompts. Therefore, our multi-level disentanglement indeed helps each trait embedding to keep their individual semantic over prompts.

4.4.2 Ablation of each component

We have conducted an ablation study to assess the impact of each component in MLPAS. For the construction of token-based instruction, three components can be adjusted: (1) essay prompt embedding; (2) prompt ID, and (3) trait tokens. In addition, regarding the multi-level disentanglement, we can control (4) the use of the prompt-wise projection. Table 10 summarizes the results when each one of the four components is removed from the final model. The ablation results indicate that all components are essential for the best performance in the evaluation of essays.

Specifically, the most significant performance drop, on average, occurs when essay prompts are excluded. These results demonstrate the importance of using the semantic information contained in the prompts. Furthermore, our observations indicate that our methods are particularly effective in leveraging the diverse semantic information contained in multiple prompts more efficiently.

4.4.3 Impact of trait ordering

The sensitivity to the order of traits in the input is a concern in the encoder-decoder ArTS (Do et al. 2024), due to its auto-regressive decoding. However, with the task replaced by a simple regression, MLPAS alleviates the impact of the order on the predicted essay scores. Figure 8 supports this well by comparing the absolute gap of QWK scores before and after reversing the order of traits in the input. The smaller gaps on the ArTS at both levels demonstrate that MLPAS is more robust to the trait order than the ArTS.

5 Conclusion

In this work, we propose a multi-level disentanglement framework named MLPAS for essay scoring. This framework achieves superior performance by leveraging token-based instruction and disentanglement at the prompt and trait levels, which together allow the model to capture distinct semantic representations for each trait and prompt. This design mitigates interference not only between prompts but also between traits, leading to improved generalization to unseen prompt-trait combinations. These insights underscore the importance of specialized modules in multi-trait, multi-prompt essay scoring. Through this, MLPAS not only outperforms state-of-the-art models in accuracy, but also offers greater efficiency with reduced latency and a smaller parameters. In particular, MLPAS demonstrates strong generalization to unseen prompts through a simple yet effective prompt-wise normalization strat-

Table 8 Prompt-wise agreement with human scores on ASAP/ASAP++ dataset when using different sizes of T5 backbones

Model	# Parameters	P1	P2	P3	P4	P5	P6	P7	P8	AVG↑(SD↓)
ArTS (T5-small)	60 M	0.696	0.669	0.682	0.732	0.712	0.743	0.712	0.492	0.680 (±0.029)
ArTS (T5-base)	220 M	0.708	0.706	0.704	0.767	0.723	0.776	0.749	0.603	0.717 (±0.025)
ArTS (T5-large)	770 M	0.701	0.698	0.705	0.766	0.725	0.773	0.743	0.635	0.718 (±0.030)
MLPAS(T5-small)	37 M	0.710	0.702	0.694	0.758	0.725	0.756	0.728	0.669	0.718 (±0.009)
MLPAS(T5-base)	113 M	0.718	0.726	0.703	0.771	0.727	0.765	0.753	0.683	0.731 (±0.007)
MLPAS(T5-large)	342 M	0.723	0.731	0.707	0.774	0.726	0.771	0.750	0.693	0.734 (±0.010)

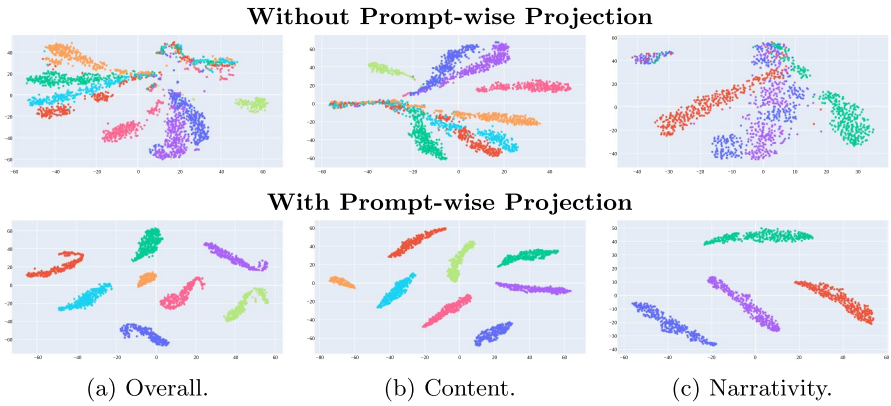

Fig. 7 T-SNE visualization of the traits with the same name across different prompts, with and without the prompt-wise projection of MLPAS, Trait embeddings just before regression are used, with each prompt color coded

egy, outperforming current cross-prompt baselines. Moreover, our findings reveal the limitations of the LLM-based few-shot essay scoring, confirming the need for lightweight, fine-tuned models for educational assessment tasks. This study offers insights into multi-level disentanglement strategies that are applicable to data with complex, multi-dimensional structures, suggesting broad potential beyond essay scoring, including applications in diverse domains.

While our model demonstrates significant advancements, several limitations should be noted. First, our experiments are limited to publicly available benchmark datasets, and the adaptability of the model to real-world settings with broader and more diverse essay topics remains untested. Second, we did not examine the model' s ability to handle the continuous introduction of new prompts over time. In practical deployments, essay prompts may evolve dynamically, necessitating models that can adapt incrementally without extensive retraining. Third, our study does not thoroughly address performance under data-scarce conditions. In real-world educational contexts, certain prompts or traits may have limited training data, which poses challenges for robustness and generalization.

Future research should therefore explore strategies for continual learning and domain adaptation to handle evolving prompts in real-time settings. In addition, developing methods to improve performance in low-resource scenarios, such as

Table 9 Trait-wise agreement with human ratings on the Feedback Prize dataset

Model	Conv	Gram	Phr	Voc	Syn	Coh	AVG↑(SD↓)
HISK	0.279	0.279	0.270	0.261	0.252	0.241	0.264 (±0.012)
STL	0.544	0.440	0.534	0.515	0.518	0.459	0.502 (±0.024)
MTL-BiLSTM	0.527	0.484	0.505	**0.519**	0.507	0.462	0.501 (±0.027)
ArTs	0.521	0.479	0.512	0.454	0.497	0.467	0.488 (±0.027)
MLPAS	**0.598**	**0.591**	**0.575**	0.518	**0.575**	**0.534**	**0.565** (±0.029)

Conv: Conventions, Gram: Grammar, Phr: Phraseology, Voc: Vocabulary, Syn: Syntax, Coh: Cohesion

Table 10 Ablation study results based on traits

Model	Overall	Content	PA	Lang	Nar	Org	Conv	WC	SF	Style	Voice	AVG↑(SD↓)
MLPAS	**0.771**	**0.746**	0.751	0.691	**0.732**	**0.710**	**0.701**	**0.701**	**0.699**	**0.702**	**0.650**	**0.714** (±0.007)
(1) w/o essay prompt	0.759	0.735	**0.755**	0.690	0.728	0.690	0.698	0.695	0.689	0.693	0.622	0.705 (±0.004)
(2) w/o prompt ID token	0.768	0.740	**0.755**	0.692	0.727	0.700	0.694	0.698	0.692	0.698	0.607	0.706 (±0.009)
(3) w/o trait tokens	0.770	0.741	0.752	**0.695**	**0.732**	0.692	0.693	**0.701**	0.690	0.683	0.633	0.707 (±0.005)
(4) w/o prompt-wise projection	0.760	0.734	0.754	0.682	0.727	0.696	0.696	0.703	0.697	0.694	0.635	0.707 (±0.009)

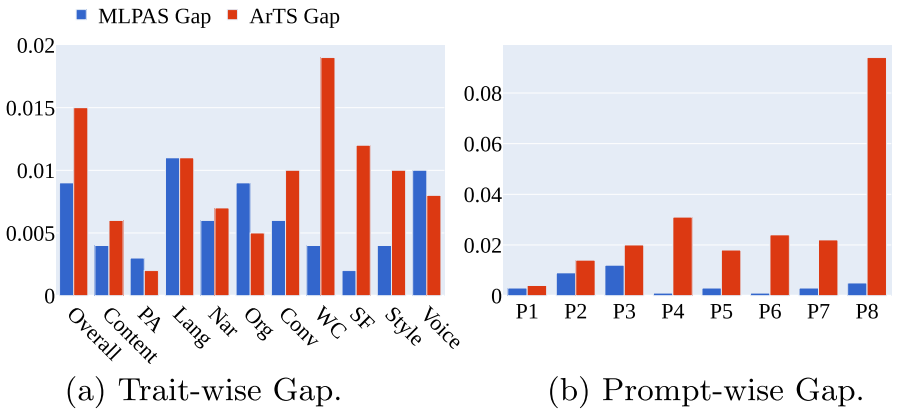

Fig. 8 Absolute QWK score gaps before and after reversing the order of traits in the input

data augmentation, prompt-aware pretraining, or meta-learning techniques, would greatly enhance the practicality of the model. Lastly, while LLMs offer powerful semantic understanding, this work highlights the importance of developing efficient task-specific models that leverage LLM capabilities without incurring their full computational overhead. Advancing toward scalable, efficient, and generalizable AES systems remains a critical direction for the next generation of educational AI.

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Authors and Affiliations

Donghee Han[1] · Daeyoung Roh[1] · Euihwan Han[2] · Hwanjun Song[3] · Mun Yong Yi[3]

✉ Hwanjun Song
songhwanjun@kaist.ac.kr

✉ Mun Yong Yi
munyi@kaist.ac.kr

Donghee Han
handonghee@kaist.ac.kr

Daeyoung Roh
dybroh@kaist.ac.kr

Euihwan Han
euihwan.han@kt.com

[1] Graduate School of Data Science, Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea

[2] KT, Korea Telecom Research, Seoul, Republic of Korea

[3] Department of Industrial and Systems Engineering, Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea